Foundations of Clinical Trichology

VOLUME 1

POSTGRADUATE CERTIFICATE IN CLINICAL TRICHOLOGY AND HAIR RESTORATION MEDICINE

Hair Longevity Institute (HLI) | International Institute of Hair Restoration (IIIOHR)

This volume is the first core teaching text for the academy. It is designed for medical doctors and structured as a postgraduate manual that can be delivered both as a written training resource and as the source document for the IIIOHR online learning platform. Study each module in sequence. The text is written to develop diagnostic reasoning, scientific understanding, and professional judgment rather than memorisation alone.

How to Use This Volume

Study Approach

Work through each module in the order presented. The architecture of this text is intentionally sequential — later modules assume mastery of earlier concepts. Each lesson is written to develop clinical reasoning frameworks rather than to support rote memorisation.

The primary aim is to train physicians to think in differential diagnosis, evidence tiers, and longitudinal management — not merely to recall treatment names.

Platform Integration

Each module includes lesson-level structure designed for ingestion into the IIIOHR online academy platform. Text blocks, tables, clinical reasoning callouts, and assessment prompts are formatted to translate directly into interactive digital learning units.

Assessment Style

End-of-module prompts require application and synthesis. Answers should reflect clinical judgment, differential thinking, and understanding of evidence quality — not isolated factual recall.

Volume Structure: Four Core Modules

This volume is organized into four foundational modules that together establish the clinical, biological, hormonal, and consultative bedrock of physician-led trichology. Each module is a prerequisite for the next.

01

Module 1

Introduction to Clinical Trichology and Hair Restoration Medicine — scope, ethics, evidence tiers, and professional responsibilities.

02

Module 2

Hair Follicle Biology and Scalp Science — embryology, anatomy, hair cycling, scalp environment, and ethnic variation.

03

Module 3

Molecular and Hormonal Regulation of Hair Growth — androgen pathways, Wnt signalling, prostaglandins, microinflammation, and treatment logic.

04

Module 4

Clinical Consultation and Hair Loss History-Taking — structured consultation framework, red flags, psychosocial considerations, and documentation standards.

MODULE 1

Introduction to Clinical Trichology and Hair Restoration Medicine

Modern clinical trichology sits at the intersection of dermatology, endocrinology, pathology, cosmetic medicine, and hair restoration surgery. A doctor-level program must therefore move beyond product-based approaches or simplified pattern-recognition alone and instead train physicians to reason clinically, stratify risk, recognise disease activity, and design longitudinal management plans.

This module establishes the intellectual and professional scaffolding upon which all subsequent modules depend. It defines what physician-led trichology means in practice, how it relates to adjacent specialties, and how evidence quality should inform clinical decision-making throughout the career of a hair medicine physician.

Module 1: Learning Outcomes

1

Define the Scope

Articulate the scope of physician-led trichology and how it differs structurally and ethically from non-medical hair practice.

2

Specialty Relationships

Explain the clinical and conceptual relationships between trichology, dermatology, endocrinology, pathology, and transplantation surgery.

3

Evidence Application

Apply evidence tiers when discussing treatments — particularly regenerative and emerging interventions — to communicate uncertainty honestly.

4

Ethical Governance

Describe the ethical, professional, and medico-legal principles that govern all aspects of hair medicine practice.

1.1 The Scope of Modern Trichology

Clinical trichology is best understood as applied hair and scalp medicine. In a physician-led setting, the task is not simply to identify "hair loss"; it is to determine whether the patient has a physiologic shedding state, a patterned miniaturisation disorder, an inflammatory process, a scarring alopecia, a shaft fragility disorder, or a systemic disease manifesting through the hair and scalp. This distinction matters because the consequences of missing early inflammatory or cicatricial disease can be permanent.

The modern field also includes longevity thinking. A patient with androgenetic alopecia (AGA) is not managed by one intervention in isolation. The physician must consider disease stage, donor stability, age-related progression, reproductive considerations, psychosocial burden, and the appropriateness of medical, regenerative, and surgical sequencing over years rather than weeks.

> **Clinical Reasoning Focus:** A good trichologist does not jump from "patterned thinning" to treatment. The first task is to identify the dominant process, the modifying factors, and the time-course of change. Only after this analysis is complete should treatment selection begin.

This scope extends to patients across the full age and sex spectrum. Hair disorders present in pediatric populations, in perimenopausal women with complex endocrine backgrounds, in post-transplant patients with ongoing androgenetic progression, and in elderly individuals with senescent thinning superimposed on earlier pattern disease. Each presentation demands a different analytical framework even when the chief complaint sounds identical.

1.2 Trichology, Dermatology, and Hair Transplantation

Dermatology remains the reference specialty for inflammatory scalp disease, complex alopecias, interface dermatitis, and clinicopathologic correlation. A doctor-level trichology curriculum should not pretend to replace dermatology training; instead, it should bring physicians substantially closer to dermatology-grade competence within the domain of hair and scalp disease. This means emphasis on morphology, trichoscopy, investigations, differentials, biopsy logic, treatment escalation, and referral boundaries.

Hair transplantation, by contrast, is a procedural discipline layered on top of medical hair science. Surgical skill without medical judgment produces poor long-term outcomes. A surgeon who transplants into an active inflammatory or cicatricial process may accelerate follicular destruction rather than restore density. Transplantation should therefore be taught as a downstream decision within a broader disease-management model, not as an autonomous aesthetic procedure divorced from diagnosis.

Dermatology Interface

- Inflammatory scalp disease diagnosis
- Clinicopathologic correlation
- Interface dermatitis classification
- Biopsy indication and interpretation
- Treatment escalation and referral

Transplantation Interface

- Surgical planning grounded in medical diagnosis
- Donor assessment and long-term stability
- Excluding active disease before procedure
- Managing medical therapy peri-operatively
- Post-surgical monitoring for progression

1.3 The Physician's Responsibilities in Hair Medicine

Hair medicine places specific and non-negotiable responsibilities on the physician. These responsibilities are not simply technical competencies — they are professional obligations that govern how the physician interacts with patients, interprets evidence, makes decisions under uncertainty, and navigates the ethical complexity of a specialty that sits within both medicine and cosmetic practice.

→ Diagnostic Precision

Recognise when hair loss is benign, self-limited, progressive, inflammatory, or scarring. Avoid premature closure on common diagnoses when the clinical picture is atypical.

→ Differential Formulation

Formulate a differential diagnosis based on onset, distribution, symptoms, and examination findings before selecting investigations or initiating treatment.

→ Selective Investigation

Use investigations selectively rather than reflexively. An extensive panel does not substitute for clinical reasoning and may generate misleading results that delay appropriate management.

→ Trichoscopy and Biopsy Logic

Interpret trichoscopy findings systematically and identify when scalp biopsy is required to confirm diagnosis, differentiate non-scarring from scarring disease, or document inflammatory activity before irreversible damage occurs.

→ Ethical Surgical Gatekeeping

Identify when surgery is inappropriate or premature — including unstabilised patterned disease, active inflammation, unrealistic expectations, or insufficient donor reserve — and document the clinical basis for deferral.

1.4 Evidence Tiers in Hair Medicine

All teaching within this academy is framed within a four-tier evidence classification. This system allows clinicians to communicate clearly about the certainty of any intervention, to distinguish established from investigational practice, and to explain treatment rationale to patients with appropriate transparency. Every treatment discussed in this curriculum is assigned an evidence tier, and clinicians should habitually think in these terms when formulating management plans and obtaining informed consent.

Tier	Meaning	Representative Examples
Tier 1	Established or widely accepted medical practice with stronger supporting data	Topical minoxidil; finasteride in appropriate male AGA; standard diagnostic workup including serum ferritin, thyroid function, and full blood count
Tier 2	Consensus-based or evolving clinical practice with growing evidence but less standardisation	Low-dose oral minoxidil; selected anti-androgen pathways in women including spironolactone and cyproterone acetate
Tier 3	Emerging or adjunctive practice where biological signal exists but heterogeneity is high	PRP protocols with variable preparation methods; selected device-based adjuncts including low-level laser therapy
Tier 4	Investigational or academy-signature concepts: biologically plausible but not yet guideline-level standard	Exosome-based therapy; partial transection and regenerative donor concepts; follicular stem-cell activation approaches

The assignment of a tier is not static. As new controlled data emerge, therapies may be reclassified upward or downward. The physician's responsibility is to track this evolution and update patient counselling accordingly.

1.5 Ethics, Consent, and Scope of Practice

Hair loss is often medically benign yet psychologically profound. This combination makes the field unusually vulnerable to overpromising, overselling, and blurring of boundaries between evidence-based medicine and aspirational marketing. The physician must actively protect against this by obtaining genuine informed consent, documenting uncertainty, distinguishing maintenance from cure, and avoiding claims that outrun the data.

Informed consent in hair medicine requires more than a signature. It requires that the patient understands the diagnosis with appropriate certainty, the natural history of their condition without treatment, the realistic outcomes and timeframes of proposed interventions, the maintenance requirements, the side-effect profiles, and the limitations of current evidence. This is especially important in Tier 3 and Tier 4 interventions, where the evidence base is evolving and outcomes are less predictable.

> **Red Flags Requiring Caution or Referral:** Pain, burning, perifollicular scale, pustulation, loss of follicular openings, rapid diffuse shedding with systemic symptoms, patchy loss with inflammation, and suspected scarring disease should all raise the threshold for reassurance-only management. These features mandate further investigation, specialist input, or both — reassurance without workup in these settings carries medico-legal and clinical risk.

The medico-legal dimension of hair medicine is underappreciated. Documenting the basis for diagnosis, the evidence tier of treatments discussed, the alternatives considered and declined, and the patient's understanding at the time of consent protects both parties in the event of a poor outcome or adverse reaction. In vulnerable cosmetic populations — particularly younger patients with early AGA and significant psychological distress — the threshold for detailed documentation should be especially high.

MODULE 2

Hair Follicle Biology and Scalp Science

Strong hair medicine begins with follicular biology. Every treatment, every disease pattern, and every surgical decision sits on top of the architecture and signalling behaviour of the follicle and its surrounding scalp environment. A physician who cannot reason from biological first principles will struggle to explain why certain therapies work, why some patients progress despite treatment, and why scarring disease demands urgency that androgenetic alopecia alone does not.

This module builds a complete biological model of the follicle — from its embryologic origins to its mature cycling behaviour, scalp microenvironment, and the clinical consequences of aging and ethnic variation. This is not background knowledge to be skimmed; it is the explanatory infrastructure for every clinical and procedural decision that follows.

Module 2: Learning Outcomes

1

Anatomy and Development

Describe the anatomy and embryologic development of the follicle and scalp, including the signalling pathways governing morphogenesis.

2

Functional Significance

Explain the functional significance of the dermal papilla, bulge region, and stem-cell niches in normal cycling and in disease states.

3

Cycle and Disease

Relate hair cycling biology — including anagen, catagen, telogen, kenogen, and exogen — to clinical disorders such as AGA, telogen effluvium, and scarring alopecias.

4

Variation and Environment

Discuss ethnic hair variation, shaft properties, and scalp microenvironmental factors as they relate to diagnosis, treatment selection, and procedural planning.

2.1 Embryology and Follicle Formation

Hair follicles arise through reciprocal signalling between epidermal keratinocytes and underlying mesenchyme. This is not a simple top-down inductive process but an iterative molecular dialogue in which epithelial and mesenchymal compartments each respond to and reinforce signals from the other. Understanding this early biology matters clinically because mature follicle behaviour — including cycling, response to injury, and regenerative potential — still depends on the integrity of epithelial-mesenchymal communication.

The developmental sequence proceeds through several morphologically distinct stages. Placode formation occurs as a focal thickening of surface ectoderm in response to Wnt-mediated pre-patterning signals. The placode then elongates downward as a hair germ, while a condensate of mesenchymal cells forms beneath — this mesenchymal condensate will become the dermal papilla. As downgrowth continues through the peg and bulbous peg stages, the follicle acquires its mature compartmental organisation.

1

Placode Formation

Wnt-mediated focal thickening of surface ectoderm; pre-patterning signal establishes follicle spacing

2

Hair Germ

Epithelial downgrowth begins; mesenchymal condensate forms beneath — the future dermal papilla

3

Hair Peg

Continued elongation; Sonic hedgehog and BMP modulation refine compartmental identity

4

Bulbous Peg

Full follicle architecture emerges; sebaceous and arrector pili attachments established

The principal signalling networks governing folliculogenesis include Wnt/beta-catenin (placode initiation and stem-cell maintenance), Sonic hedgehog (matrix proliferation and lower follicle identity), bone morphogenetic protein modulation (spatial patterning and inhibitory boundary setting), and fibroblast growth factor networks (branching and mesenchymal differentiation). For the clinician, this molecular groundwork explains why mature follicles are not simply passive tubes but dynamic signalling structures — and why regenerative therapies seeking to reactivate growth are, in essence, attempting to recapitulate developmental biology in an aged or diseased tissue context.

2.2 Follicle Anatomy in Depth

Anatomical Compartments

The mature follicle is divided into three main regions defined by reference landmarks. The **infundibulum** extends from the ostium (follicular opening at the skin surface) down to the level of the sebaceous duct. This region is in continuity with the surface epithelium and shares its barrier properties. The **isthmus** continues from the sebaceous duct to the insertion of the arrector pili muscle; within this zone lies the bulge — a critical stem-cell niche discussed separately. The **lower follicle** encompasses the stem, suprabulbar region, and the bulb, and is present only during active growth phases.

Inner and Outer Root Sheath

The outer root sheath (ORS) is contiguous with the surface epidermis and provides a structural sleeve around the growing follicle. It harbours several progenitor populations and is the origin of many cells involved in wound healing and follicular regeneration. The inner root sheath (IRS) is a transient but functionally essential structure that shapes and guides the growing hair shaft as it ascends through the follicular canal. The IRS is completely resorbed at the level of the isthmus and does not extend to the surface.

The Dermal Papilla

The dermal papilla (DP) is the mesenchymal signalling centre of the follicle. Residing at the base of the bulb, it is enclosed by matrix keratinocytes, which proliferate and differentiate to generate the hair shaft and IRS. The DP is not a passive support structure — it actively regulates anagen initiation, maintenance, shaft calibre, and cycling behaviour through secretion of growth factors, morphogens, and paracrine signals.

DP cell number correlates with fibre calibre. In AGA, miniaturisation is associated with reduction in DP volume and altered DP cell signalling behaviour, not merely reduced androgen availability.

Matrix Keratinocytes

These are among the most rapidly dividing cells in the human body during anagen. Their proliferative output determines fibre elongation rate. Disruption of matrix proliferation — by drugs, nutritional deficiency, inflammation, or systemic illness — results in shaft defects, anagen effluvium, or cycle truncation.

2.3 Bulge Stem Cells and Regenerative Potential

The bulge region, located within the isthmus at the level of the arrector pili insertion, contains a population of epithelial stem cells (EpSCs) that are relatively quiescent under homeostatic conditions. These cells express key markers including Lgr5, CD34, and Keratin 15, and they possess the capacity for self-renewal and multipotency, being able to contribute to follicular cycling, sebaceous gland renewal, and epidermal repair following injury.

During the hair cycle, activating signals — principally from the dermal papilla and dermal sheath — mobilise a subset of these cells to generate the transit-amplifying matrix population responsible for anagen initiation. The precision of this activation, and the capacity of stem cells to respond, determines whether each new cycle produces a fibre of equivalent calibre to the last, or whether a subtle decline begins.

> **Why This Matters Clinically:** Non-scarring alopecia usually preserves the bulge architecture needed for potential regrowth. Cicatricial (scarring) alopecia destroys the bulge stem-cell niche, resulting in permanent follicular loss. This is the core biological reason that early diagnosis of inflammatory and scarring disease changes prognosis: once the bulge is destroyed, no current therapy can restore the follicle. Speed of diagnosis is equivalent to preservation of regenerative capacity.

In regenerative hair medicine, many therapeutic concepts — including PRP, exosome delivery, microneedling with growth factor application, and stem-cell-derived conditioned media — are, to varying degrees, attempting to modulate the signalling environment of the bulge and perifollicular niche. Understanding bulge biology allows the physician to evaluate these approaches with appropriate scientific discrimination rather than accepting promotional claims at face value.

2.4 Hair Shaft Architecture

The mature hair shaft is a non-living composite fibre produced by the differentiation and keratinisation of matrix keratinocytes. Its structural organisation determines the mechanical, optical, and cosmetic properties that patients perceive as hair quality, density, and texture.

Cuticle

Outermost layer composed of overlapping, scale-like cells. Protects the cortex from environmental damage. Disruption by chemical processing, UV, heat, or mechanical trauma results in visible surface roughness, dullness, and increased fragility. Cuticle integrity influences light reflection and perceived shine.

Cortex

The dominant structural layer. Contains macro- and microfibrils of alpha-keratin embedded in a matrix of matrix proteins. The cortex determines tensile strength, elasticity, curl behaviour, colour (via melanin granule distribution), and most of what patients and clinicians perceive as fibre "fullness" or calibre.

Medulla

Present in some fibre types and ethnic groups; may be absent in fine fibres. Its functional role is not fully characterised, but medullary presence correlates with coarser fibre types. Absence of a medulla is normal and does not indicate pathology.

Cross-sectional shape, curvature, and follicular emergence angle vary significantly by ethnicity. Straight fibres (common in East Asian populations) emerge at a relatively perpendicular angle and have a round cross-section. Tightly coiled fibres (common in African-origin populations) have an elliptical cross-section and a curved subcutaneous follicular canal. These variations have direct clinical implications: breakage patterns differ, FUE punch selection must account for curvature, implantation angle must match natural emergence, and traction alopecia patterns differ by hair type and grooming practice.

2.5 Hair Cycle Biology

The hair cycle is the fundamental biological rhythm of the follicle. Misunderstanding the cycle leads to misinterpretation of clinical findings and patient complaints. Every major hair disorder can be mapped to a disruption of cycling behaviour — either a pathologic shift in cycle phase proportions, truncation of anagen, or failure of successful anagen re-entry.

Phase	Core Biology	Approximate Clinical Meaning	Disorders Showing Disturbance
Anagen	Active growth; matrix keratinocyte proliferation; shaft elongation	Determines fibre length, calibre, and ultimate visible density	Shortened in AGA; disrupted in anagen effluvium from chemotherapy or alopecia areata
Catagen	Controlled involution; apoptosis-driven regression of the lower follicle; dermal papilla condenses upward	Brief transition state; lasts approximately 2–3 weeks	Rarely clinically dominant alone; may be prolonged in certain inflammatory states
Telogen	Relative resting state; club hair retained; follicle arrested above dermal papilla	Sets up later physiologic shedding of the club hair	Expanded proportion in telogen effluvium; contributes to apparent density reduction
Exogen	Active shedding of the club hair, now recognised as a regulated process	Normal hair fall of 50–100 hairs/day in a healthy scalp	Dysregulated in some inflammatory states; contributes to acute shedding complaints
Kenogen	Empty follicle interval between club hair release and new anagen entry	Extends the period before new fibre emerges; worsens perceived density loss	Increasingly important in patterned thinning; prolonged kenogen reduces apparent density disproportionately

For scalp follicles, anagen may last two to six years or longer, which is why scalp hair can grow to considerable length. The proportion of follicles in anagen at any given time (the anagen-to-telogen ratio) normally favours anagen by approximately 85–90%. A shift of even 10–15% of follicles into telogen will generate clinically noticeable shedding. In AGA, the primary problem is not just increased telogen proportion but progressive shortening of successive anagen cycles — each new fibre is produced for less time than the last, resulting in progressively finer and shorter fibres over years.

2.6 The Scalp as a Biologic Environment

The scalp is not simply the surface on which follicles sit. It is a specialised, dynamic biologic environment that profoundly influences follicular performance, treatment outcomes, and disease susceptibility. A physician who treats follicles in isolation — ignoring the surrounding tissue environment — will systematically underperform in both medical management and procedural planning.

The scalp possesses some of the highest follicular density in the body, a rich dermal vasculature that supports the metabolic demands of active anagen follicles, prominent sebaceous activity that influences surface lipid composition and microbiome ecology, and a mechanically exposed surface subject to UV radiation, traction, chemical exposure, occlusion, and grooming trauma. Each of these factors contributes to the follicular microenvironment in ways that can be protective or injurious depending on context.

Scalp Microbiome

Malassezia species are integral components of scalp ecology. Dysbiosis — shifts in Malassezia burden or diversity — is associated with seborrhoeic dermatitis and may contribute to perifollicular inflammatory load. Microbiome research is an active area with growing clinical relevance.

Vascular Supply

Scalp perfusion supports matrix keratinocyte metabolism. Evidence for vascular support as a therapeutic target underpins the use of vasodilatory agents such as minoxidil and the rationale for some microneedling and laser protocols.

Barrier Function

Scalp barrier integrity influences inflammatory susceptibility, product penetration, and infection risk. Barrier compromise — from over-washing, harsh chemical processing, or underlying dermatitis — amplifies follicular vulnerability.

Sebum Dynamics

Sebum composition influences surface pH, antifungal defence, and epidermal hydration. Androgens upregulate sebaceous activity, linking sebaceous gland biology directly to the androgenetic pathway relevant in AGA.

2.7 Ethnic Hair Variation and Its Clinical Relevance

Clinicians practising hair medicine across diverse populations must understand that "hair" is not a biologically uniform entity. Straight, wavy, tightly curled, and coiled hair types differ significantly in shaft morphology, follicular geometry, breakage pattern, sebum distribution, and the pattern of disorders to which they are susceptible. These differences are not cosmetic technicalities — they have direct clinical, surgical, and diagnostic implications.

East Asian Hair (Typically Straight)

Circular cross-section; relatively perpendicular emergence angle; high tensile strength; less susceptibility to breakage under normal grooming. FUE extraction is technically straightforward due to minimal follicular curvature. Miniaturisation in AGA may appear less dramatic per follicle unit due to initially larger calibre.

European Hair (Typically Wavy to Straight)

Oval cross-section; moderate emergence angle; variable calibre. AGA presents with the classical Norwood distribution. Female pattern loss tends to follow Ludwig classification. Seborrhoeic dermatitis is common. Surgical extraction and implantation are technically well-standardised in this population.

Afro-textured Hair (Tightly Curled to Coiled)

Elliptical cross-section; significant follicular curvature subcutaneously; high susceptibility to traction-induced breakage and traction alopecia; increased risk of central centrifugal cicatricial alopecia (CCCA). FUE punch selection must account for curvature to avoid transection. Sebum distribution is less uniform, increasing dryness and fragility of distal shaft segments.

South Asian and Middle Eastern Hair

Variable from straight to wavy to moderately curled; often high calibre; strong genetic predisposition to AGA in many lineages. Chemical and heat processing common. Traction alopecia at frontotemporal hairline from certain traditional hair practices may coexist with AGA and must be distinguished on examination.

2.8 Aging and Follicular Senescence

Aging imposes a progressive and multidimensional deterioration of follicular performance that is distinct from — though often superimposed on — genetic patterned hair loss. Understanding follicular aging allows the clinician to explain to patients why treatment response may differ at different life stages, why new procedures do not simply "reset" the biological clock, and why preventive strategies introduced early in life offer more biological capital than late-stage rescue.

Key mechanisms of follicular aging include: decline in bulge stem-cell responsiveness and signalling fidelity; accumulation of senescent fibroblasts in the perifollicular dermis, which alter growth factor secretion profiles; progressive perifollicular extracellular matrix remodelling with collagen crosslinking and fibrosis; decreased melanocyte stem-cell activity leading to greying; and reduced efficiency of anagen re-entry following each successive telogen. Clinically, patients perceive the cumulative result of these changes as reduced scalp density, slower regrowth following shedding events, progressively finer shaft calibre, and reduced resilience to stressors that previously caused only temporary shedding.

Clinical Integration: Aging interacts with genetic susceptibility in AGA in a way that is clinically important. A patient with a moderate genetic predisposition to AGA may remain stable until the perifollicular environment accumulates sufficient age-related changes — including microfibrosis and reduced stem-cell reserve — to unmask follicular androgen sensitivity. This explains why clinically significant AGA can appear in the fifth or sixth decade in individuals with apparent early-onset family history. Treatment plans must account for the patient's biological age, not just their calendar age.

MODULE 3

Molecular and Hormonal Regulation of Hair Growth

Hair growth is controlled by interlocking signalling systems rather than by a single hormone or molecular switch. This complexity is clinically important: it explains why no single drug completely prevents AGA progression, why different patients respond differently to the same intervention, and why combination strategies are increasingly the clinical standard. Doctors must understand these pathways to interpret AGA, female hormonal hair loss, inflammatory miniaturisation, and the logic behind both established and emerging therapies.

This module moves from androgenic signalling — the most clinically familiar pathway — through the broader molecular landscape of follicular control, including Wnt/beta-catenin signalling, prostaglandin biology, growth mediator networks, microinflammatory processes, and sex-specific endocrine considerations. The module concludes by mapping this biology directly onto treatment logic, grounding clinical decisions in mechanistic understanding.

Module 3: Learning Outcomes

Androgen Signalling

Explain in mechanistic terms how androgen signalling contributes to patterned hair loss, including the role of 5-alpha reductase, DHT, androgen receptor binding, and the downstream gene expression changes that drive miniaturisation.

Pathway Biology

Describe the roles of Wnt/beta-catenin signalling, prostaglandin balance, growth mediators, and vascular factors in follicular cycling, maintenance, and miniaturisation.

Microinflammation and Fibrosis

Discuss microinflammation, perifollicular fibrosis, and aging as modifiers of follicular performance that operate independently of — or in synergy with — androgen-driven pathways.

Clinical Application

Apply this molecular framework directly to treatment sequencing decisions and to patient education about why treatments work, when they fail, and what realistically can and cannot be achieved.

3.1 Androgen Biology and Follicular Response

Androgens do not affect all follicles equally. This regionally selective sensitivity — the central biological paradox of AGA — determines why the frontal and vertex scalp miniaturises while occipital follicles are relatively preserved, even though systemic androgen levels are identical across all scalp regions.

In genetically susceptible scalp regions, testosterone is converted to dihydrotestosterone (DHT) by 5-alpha reductase isoenzymes — primarily Type II within the dermal papilla. DHT binds the androgen receptor (AR) with approximately five times greater affinity than testosterone and has a substantially slower dissociation rate. The resulting DHT-AR complex translocates to the nucleus and alters gene expression within the dermal papilla cell, progressively shifting the secretory profile toward signals that shorten anagen, reduce shaft calibre, and favour miniaturisation over successive cycles.

Key Molecular Steps

- Testosterone enters susceptible dermal papilla cell
- 5-alpha reductase Type II converts testosterone → DHT
- DHT binds androgen receptor with high affinity
- DHT-AR complex enters nucleus
- Altered gene expression → shifted DP secretory profile
- Progressive anagen shortening and calibre reduction

A Critical Clinical Point

Androgenic signalling alone does not explain all clinical behaviour in patterned hair loss. Many patients with entirely normal circulating androgen levels — both male and female — develop AGA, because what determines follicular vulnerability is local androgen receptor density, 5-alpha reductase isoenzyme expression levels, intracrine androgen metabolism within the dermal papilla, and the pre-existing genetic programming of individual follicular units rather than serum androgen excess.

This is why serum testosterone measurement alone is insufficient and often misleading in AGA workup, and why DHT measurement is not routinely clinically useful in guiding treatment decisions in standard male AGA.

3.2 Androgen Receptor Genetics and Age of Activation

AGA is a polygenic, complex-trait disorder. Genome-wide association studies have identified numerous susceptibility loci, but the most clinically important genetic signal maps to the androgen receptor gene on chromosome Xq11-12. Since AR is X-linked, maternal inheritance was historically emphasised — hence the common but oversimplified instruction to "look at the mother's father." In reality, both maternal and paternal lineages contribute to AGA risk through multiple loci, and family history should be interpreted probabilistically rather than deterministically.

A clinically useful conceptual model is that susceptible follicles are not uniformly "active" from puberty onward; rather, they appear to become vulnerable once a genetically programmed threshold of age, hormonal milieu, and local follicular signalling is reached. This threshold varies substantially between individuals even within the same family, which explains why one sibling progresses in the early twenties while another with the same family history remains relatively stable until the fourth decade.

> **Teaching Pearl for Patient Education:** Genes do not merely determine whether a patient will lose hair — they also govern when follicular androgen sensitivity becomes clinically active. This distinction matters for patient counselling: early onset may indicate a lower threshold of activation, a broader zone of susceptibility, or a higher receptor density, all of which have implications for prognosis and the urgency of preventive medical therapy.

Age of onset carries prognostic information. Patients presenting with AGA in the late teens or early twenties tend to have more aggressive progression, wider zones of susceptibility, and a greater long-term requirement for sustained medical therapy. They also represent the group at highest risk of premature surgical intervention before progression has stabilised — one of the most common and preventable sources of poor hair restoration outcomes.

3.3 Wnt/Beta-Catenin Signalling

Wnt/beta-catenin signalling occupies a central position in follicular biology at every level — from embryologic development through mature cycling behaviour. During development, Wnt signals from the epidermis initiate placode formation and regulate morphogenesis. In the adult follicle, Wnt activity is required for anagen initiation, dermal papilla cell identity maintenance, and bulge stem-cell activation. Suppression of this pathway is consistently associated with impaired regenerative activity, failure of anagen re-entry, and miniaturisation.

In the canonical pathway, Wnt ligand binding to Frizzled receptors and LRP5/6 co-receptors inhibits the destruction complex — the kinase machinery that otherwise phosphorylates and targets beta-catenin for proteasomal degradation. When the pathway is active, beta-catenin accumulates in the cytoplasm and translocates to the nucleus where it complexes with TCF/LEF transcription factors and drives expression of Wnt target genes including cyclin D1 and AXIN2. Disruption of this process — by pathway antagonists such as DKK-1, by aging-related changes in co-receptor expression, or by inflammatory mediators — reduces anagen induction capacity.

Clinical and Therapeutic Significance: Many regenerative or investigational interventions — including certain formulations of PRP, Wnt pathway agonists under investigation, and scalp microneedling with growth factor co-delivery — are, directly or indirectly, attempting to support pathways associated with Wnt activation and anagen re-entry. Understanding this molecular target allows the physician to evaluate emerging therapies with mechanistic specificity rather than relying solely on clinical trial outcomes that may not be available for years.

3.4 Prostaglandins and Cycling Behaviour

Prostaglandin signalling represents an underappreciated but increasingly important dimension of hair cycle regulation. Within follicular biology, prostaglandins act as paracrine regulators of cycling state and growth behaviour, and their balance within the scalp environment appears to shift meaningfully in AGA and other conditions affecting follicular output.

Prostaglandin D2 (PGD2) — Inhibitory

PGD2, acting through the CRTH2 (DP2) receptor, has been implicated as an inhibitory signal in patterned hair loss. Studies have demonstrated elevated PGD2 concentrations in balding scalp regions compared to non-balding occipital regions in AGA subjects. Downstream signalling via CRTH2 on hair follicle cells appears to suppress anagen induction and reduce follicular responsiveness to activating signals. This finding reinforced the concept that AGA involves a broader shift in the follicular signalling environment — not simply androgen excess.

Prostaglandin E2 (PGE2) and PGF2α — Potentially Supportive

Other prostaglandins, including PGE2 and the prostamide/prostaglandin F2α class, appear to support follicular growth — a finding with direct therapeutic relevance. Bimatoprost, a prostamide analogue used topically, was found to promote eyelash and eyebrow growth, and this has prompted investigation of prostaglandin analogues in scalp hair restoration. While this area has not yet yielded a standard-of-care scalp treatment, it illustrates the mechanistic depth of prostaglandin biology in hair medicine.

For the clinician, the prostaglandin axis reinforces a key conceptual principle: AGA and related patterned disorders are not simply "too much DHT acting on sensitive follicles." They represent a broader shift in the molecular environment of susceptible follicular units — a shift that involves androgenic, inflammatory, prostaglandin-mediated, and cycling regulatory components acting simultaneously and interactively. This complexity is why no single intervention completely halts progression in all patients and why multi-pathway thinking is increasingly the standard of care.

3.5 Growth Mediators and Vascular Factors

Multiple growth mediators and vascular factors regulate follicular proliferation, involution, perifollicular matrix remodelling, and the nutritive support upon which anagen depends. A working knowledge of these mediators is essential for understanding both the rationale and the limitations of regenerative therapies in clinical practice.

IGF-1 (Insulin-like Growth Factor-1)

Promotes matrix keratinocyte proliferation and supports anagen maintenance. IGF-1 signalling in the dermal papilla is a critical component of the DP's anagen-promoting secretory profile. Nutritional states, insulin resistance, and growth hormone axis function all influence available IGF-1.

VEGF (Vascular Endothelial Growth Factor)

Promotes perifollicular angiogenesis and ensures adequate perfusion of the metabolically demanding anagen bulb. VEGF is expressed in the outer root sheath and dermal papilla. Reduced VEGF expression or perifollicular vascular density is observed in miniaturising follicles. Minoxidil's mechanism includes upregulation of perifollicular VEGF expression.

FGFs (Fibroblast Growth Factors)

Multiple FGF family members influence follicular biology. FGF-7 (KGF) and FGF-10 promote epithelial proliferation and anagen induction. FGF-5 has been identified as a catagen-promoting factor; mice lacking FGF-5 exhibit prolonged anagen and "angora" phenotype. FGF5 antagonism is under investigation as a target for anagen extension.

TGF-β (Transforming Growth Factor-Beta)

Acts as a key catagen-inducing and growth-suppressive signal. Elevated TGF-β1 and TGF-β2 within miniaturising follicles and perifollicular fibrotic niches contribute to anagen suppression, premature catagen entry, and progressive fibrosis. TGF-β signalling connects androgen biology, microinflammation, and fibrotic deterioration in a mechanistically unified model of AGA.

PRP and other regenerative approaches are frequently justified by their capacity to concentrate or modulate growth factor exposure at the follicular level. The theoretical appeal is sound: PRP preparations contain supraphysiologic concentrations of PDGF, TGF-β1, VEGF, and IGF-1, which together could support follicular activity. However, real-world outcomes vary substantially according to preparation protocol (centrifugation speed, platelet yield, activation method), disease stage, patient selection, injection technique, and frequency. Physicians should communicate this heterogeneity to patients rather than presenting PRP as a standardised, predictable intervention.

3.6 Microinflammation and Perifollicular Fibrosis

Perifollicular microinflammation is increasingly recognised as a consistent and potentially pathogenic feature of androgenetic alopecia, rather than merely an incidental finding. Histopathologic studies have demonstrated perifollicular lymphohistiocytic infiltrates in a significant proportion of AGA biopsies — even in early-stage disease and in patients who do not report scalp symptoms. This finding has shifted the conceptual model of AGA from a purely androgen-driven miniaturisation process to one in which inflammatory and fibrotic mechanisms contribute actively to follicular decline.

The relationship between androgens and perifollicular inflammation is not fully characterised but appears bidirectional. Androgenic stimulation of sebaceous activity may promote Malassezia-driven inflammatory signalling. Sebum composition may alter scalp surface chemistry in ways that activate pattern recognition receptors. Perifollicular mast cells and macrophages release cytokines and matrix metalloproteinases that remodel the perifollicular connective tissue sheath, progressively replacing functional extracellular matrix with collagen-rich fibrotic tissue.

Clinical Significance of Microinflammation: This concept helps explain why some patients with AGA progress more rapidly than expected, experience scalp discomfort, tenderness, or pruritus, or demonstrate perifollicular scale, subtle erythema, or follicular cast formation on trichoscopy. It also supports the imperative to assess the scalp itself — not just the hair pattern — and to consider whether anti-inflammatory components (topical or systemic) are warranted in the management plan when microinflammatory features are prominent.

Perifollicular fibrosis, once established, compounds the problem by reducing the structural and biochemical accessibility of the dermal papilla to regenerative signals, reducing the capacity of the bulge stem-cell niche to respond to activating cues, and mechanically compressing the follicular unit. In advanced AGA with significant fibrosis, the response to medical therapy — including minoxidil and anti-androgens — may be attenuated, and surgical planning must account for the possibility of fibrosis affecting graft survival at the recipient site.

3.7 Sex-Specific Endocrine Influences

Male and female pattern hair loss share androgenic biological mechanisms but differ substantially in their endocrine context, clinical presentation, differential diagnosis, and treatment approach. Female hair loss, in particular, must be taught as a broader hormonal and systemic medicine problem — not as a simple mirror image of male AGA with different drugs substituted.

Male Pattern Hair Loss

Occurs predominantly in the setting of normal male androgen physiology acting on genetically susceptible follicles. The endocrine evaluation is typically straightforward unless the clinical picture is atypical — very early onset, rapid progression, systemic virilisation symptoms, or failure to respond to adequate anti-androgen therapy. In standard male AGA, extensive hormonal testing rarely alters the management plan.

Key treatment decision: finasteride vs. dutasteride depends on degree of 5-alpha reductase Type I vs. Type II involvement, patient age, fertility considerations, and tolerance profile.

Female Pattern Hair Loss

Considerably more heterogeneous in endocrine context. Several distinct subpopulations exist:

- **No overt endocrine abnormality:** normal androgens, normal thyroid, diffuse bitemporal pattern — often true female AGA with genetic predisposition
- **Hyperandrogenism:** elevated free or total testosterone, low SHBG, evidence of PCOS, adrenal enzyme disorders
- **Perimenopausal/menopausal:** declining oestrogen reduces competitive androgen balance, unmasking relative androgenic sensitivity
- **Thyroid dysfunction:** both hypothyroidism and hyperthyroidism alter cycling and should be tested routinely
- **Mixed or compounded:** AGA pattern coexisting with active telogen effluvium from iron deficiency, thyroid disease, or nutritional deficit

The clinical implication is that female diffuse thinning or female pattern hair loss requires a systemic endocrine evaluation in every case, not just when historical features are overtly suggestive. Hormonal contributors modify treatment selection, influence prognosis, and sometimes require management in their own right before hair-specific therapy is optimised. A physician who treats female hair loss purely as a local follicular problem — without integrating the broader endocrine context — will miss treatable contributors and produce inferior outcomes.

3.8 Treatment Logic Derived from Molecular Biology

The molecular biology of hair loss is not academic ornamentation — it is the mechanistic foundation for every treatment decision in this specialty. Each major therapeutic category can be mapped directly onto a biological pathway, allowing the physician to explain mechanistic rationale to patients, predict which patients are most likely to respond, anticipate treatment limitations, and design logical combinations.

Minoxidil (Topical and Oral)

Supports anagen extension and follicular metabolic activity through multiple mechanisms including perifollicular VEGF upregulation, potassium channel opening in vascular smooth muscle, and possible direct effects on matrix keratinocyte proliferation. Minoxidil does not directly correct androgen signalling; it works around androgen-driven miniaturisation by supporting follicular activity through non-androgenic pathways. This explains its efficacy in both male and female AGA, in telogen effluvium, and even in non-androgenetic contexts such as alopecia areata adjunctively.

Finasteride and Dutasteride

Selectively inhibit 5-alpha reductase, reducing scalp DHT levels and thereby diminishing the primary androgenic drive for miniaturisation. They do not reverse every element of established miniaturisation biology — fibrosis, reduced stem-cell responsiveness, and altered DP secretory profiles persist and may not fully normalise. This is why outcomes are better in earlier disease, why stabilisation is a more reliable outcome than full regrowth, and why maintenance is lifelong.

Anti-inflammatory Scalp Care

May be relevant when microinflammation is clinically present — evidenced by pruritus, perifollicular scale, erythema, or trichoscopic features of follicular inflammation. Zinc pyrithione, ketoconazole, and clobetasol-based preparations reduce Malassezia burden and inflammatory signalling. The degree to which anti-inflammatory management independently influences miniaturisation trajectory remains under investigation but is biologically plausible.

Regenerative and Investigational Therapies

PRP, exosome preparations, and growth factor complexes are most biologically plausible as adjuncts in follicles that retain sufficient architectural integrity — viable bulge stem cells, functional DP, and preserved vascular supply. In advanced miniaturisation with fibrosis, the biological target for these therapies is compromised. They function as growth-supporting signals in a permissive environment, not as architectural restoration tools in a destroyed one. This supports their use in early-to-moderate disease as an adjunct to core medical therapy, rather than as monotherapy or as alternatives to surgery in advanced cases.

MODULE 4

Clinical Consultation and Hair Loss History-Taking

Doctors reach a higher level in hair medicine when the consultation becomes structured, efficient, and hypothesis-driven rather than reactive and descriptive. The history is not a bureaucratic preamble to examination — it is the primary diagnostic instrument of the trichology consultation. A well-constructed history narrows the differential, reveals modifying factors, identifies red flags that change urgency, and sets up the examination and investigation plan with clinical precision.

This module provides a comprehensive framework for the hair loss consultation, covering symptom categorisation, tempo analysis, endocrine and reproductive history in women, family history interpretation, drug review, trigger identification, psychosocial assessment, and documentation standards. A structured consultation template is provided at the end of the module as a practical clinical tool.

Module 4: Learning Outcomes

1

Structured Consultation

Use a structured, hypothesis-driven consultation framework across the common hair presentations encountered in a general or specialist trichology practice.

2

Symptom Differentiation

Differentiate shedding, thinning, breakage, and focal loss through history alone, before examination, using systematic questioning of onset, tempo, pattern, and associated symptoms.

3

Contributor Recognition

Recognise historical clues to endocrine, inflammatory, nutritional, iatrogenic, and psychosocial contributors, and integrate them into a working diagnostic hypothesis.

4

Documentation Standards

Document findings in a manner that supports longitudinal management, enables treatment outcome tracking, and satisfies medico-legal standards for a cosmetically sensitive clinical population.

4.1 Defining the Patient's Actual Complaint

The first and most important step in any hair consultation is precise translation of the patient's subjective complaint into a clinically defined category. Patients frequently use identical language to describe fundamentally different problems. "My hair is falling out" may describe excessive daily shedding, progressive reduction in ponytail circumference, progressive frontal recession noticed in photographs, cosmetically significant breakage, a single enlarging bald patch, or concern following a surgical procedure. Each of these describes a distinct biological process with different diagnoses, investigations, and management pathways.

Before asking about onset, tempo, or triggers, the physician should establish which of the following four primary categories best characterises the patient's complaint. These categories are not mutually exclusive — two or more may coexist — but identifying the dominant presentation determines the initial diagnostic branch.

Category 1: Increased Shedding

Patient notices excessive daily hair fall — in the shower, on the pillow, in the brush. May or may not be accompanied by visible thinning yet. Key differential: telogen effluvium (acute or chronic), AGA with prominent shedding phase, early alopecia areata, thyroid dysfunction, nutritional deficiency.

Category 2: Decreased Density

Patient notices reduced volume, decreased ponytail thickness, or scalp visibility without necessarily reporting excessive daily shedding. Often represents patterned miniaturisation (AGA or female pattern hair loss), chronic telogen effluvium, or silent progressive disease. Frequently underreported for months to years before consultation.

Category 3: Shaft Breakage

Patient notices shorter hairs, flyaways, or rough texture rather than root-shed hairs. Distinguished from true effluvium by the absence of a white club or dark anagen root on shed fibres. Associated with chemical or heat damage, traction, nutritional deficiencies affecting shaft integrity, or structural shaft disorders.

Category 4: Focal Alopecia or Visible Patches

Patient identifies specific bare or thinning areas, enlarging patches, or hairline recession. Key differential: alopecia areata, trichotillomania, traction alopecia, scarring alopecias (LPP, CCCA, FAPD), tinea capitis, or focal AGA. Asymmetry, symptoms, and scalp surface appearance narrow the differential substantially.

4.2 Onset, Tempo, and Pattern

Time-course is one of the most powerful and underutilised diagnostic tools in trichology. The trajectory of a complaint — when it started, how it began, whether it has progressed, stabilised, or fluctuated, and how rapidly change has occurred — often provides more diagnostic signal than any single clinical finding or laboratory test.

1 Acute Onset (Weeks)

Onset over 6–12 weeks suggests telogen effluvium, particularly if associated with an identifiable trigger 8–12 weeks prior. Sudden diffuse shedding with anagen hairs suggests anagen effluvium — chemotherapy, severe systemic illness, or toxic exposure.

2 Subacute (Months)

Progressive loss over 3–6 months raises suspicion for a combination process: chronic telogen effluvium superimposed on patterned loss, evolving female pattern hair loss, or early inflammatory alopecia. Nutritional and hormonal contributors are often operative in this timeframe.

3 Chronic Progressive (Years)

Slow, progressive thinning over years is the hallmark of AGA in both sexes. Patients may present having normalised gradual change until a threshold of cosmetic visibility is crossed. Progression over decades with appropriate topographic distribution is reassuring against active inflammation.

4 Episodic or Fluctuating

Recurrent episodes of shedding separated by partial recovery suggest chronic or recurrent telogen effluvium, seasonal variation, cyclical hormonal influence (e.g., post-contraceptive pill), or incompletely treated systemic disease. This pattern warrants systematic endocrine and nutritional review.

Symptoms associated with the hair loss complaint alter the differential diagnosis significantly. Burning, pain, or tenderness raises suspicion for an inflammatory or scarring process. Scalp pruritus may accompany seborrhoeic dermatitis, psoriasis, or early lichen planopilaris. Complete absence of symptoms, by contrast, is typical of AGA and non-inflammatory telogen effluvium. Stepwise progression with asymmetry, eyebrow loss, or follicular plugging should never be attributed to common patterned loss without further investigation.

4.3 Menstrual, Reproductive, and Endocrine History in Women

In women presenting with any form of hair loss — whether diffuse thinning, patterned loss, or mixed presentation — the endocrine and reproductive history is not an optional component; it is central to interpretation and management. A consultation that proceeds directly from the hair complaint to examination without this history will miss a substantial proportion of treatable contributors and produce systematically inferior clinical outcomes.

The following domains should be covered routinely in every female hair loss consultation. The depth of inquiry in each area may vary based on the initial responses, but the opening question in each domain should always be asked.

Menstrual Regularity and Dysmenorrhoea

Irregular, infrequent, or absent periods suggest anovulation and possible androgen excess in the context of PCOS or adrenal dysfunction. Dysmenorrhoea severity may indicate endometriosis, which has broader hormonal and inflammatory implications.

Fertility and Reproductive History

Difficulty conceiving or recurrent pregnancy loss may indicate PCOS, thyroid dysfunction, or hyperprolactinaemia — all of which can contribute to hair disorders. Postpartum state must always be identified as this is the most common trigger for acute telogen effluvium in women.

Contraceptive and Hormonal Medication History

Current and recent hormonal contraceptive use is essential. Androgenic progestogens (levonorgestrel, norethisterone) can worsen AGA susceptibility. Initiation of the combined pill after years of injectable progestogen can trigger a withdrawal effluvium. Cessation of the combined pill unmasks pre-existing female AGA previously suppressed by oestrogenic protection.

Acne, Hirsutism, and Weight Change

The presence of acne (particularly jawline or inflammatory acne), unwanted facial or body hair growth, and unexplained weight gain — especially with central adiposity — constitutes the classical triad of androgen excess and should trigger PCOS workup.

Menopausal Transition and Thyroid Symptoms

Perimenopausal and menopausal women may develop or worsen female pattern hair loss as oestrogen decline reduces the competitive protection against follicular androgen sensitivity. Thyroid symptoms (fatigue, cold intolerance, weight change, palpitations, bowel change) should be systematically asked about and thyroid function tested routinely — not only when symptoms are present.

4.4 Family History and Genetic Patterning

Family history of hair loss carries genuine prognostic information but must be interpreted with appropriate clinical sophistication rather than taken at face value. The most common error is to treat the family history as a binary predictor — "he has a bald father, so he will go bald" — rather than as one input into a probabilistic model that includes age, current stage, rate of progression, and modifying factors.

What to Elicit

- Both maternal and paternal family members, not only the father's side
- Age of onset in affected relatives
- Severity and topographic pattern achieved
- Whether apparent "thinning" may have represented diffuse disease, late-life senescent change, or telogen effluvium rather than true AGA
- Whether affected relatives were ever treated and what response was observed
- Female relatives with PCOS, irregular periods, or female pattern loss

Clinical Interpretation

Family history of early-onset, severe AGA in multiple first-degree relatives on both sides identifies a patient at higher genetic risk who may benefit from earlier initiation of medical therapy, more conservative surgical planning, and longer-term prognostic counselling. Family history of late-onset, mild thinning in a single relative is a much weaker predictor and should be weighted accordingly.

Importantly, family history does not determine inevitability or rate of progression. Patients with strong family histories who begin effective medical therapy early may achieve substantially better long-term outcomes than untreated individuals with less severe family pedigrees. This point is therapeutically empowering and should be communicated clearly.

4.5 Drug and Treatment History

A thorough medication history is non-negotiable in hair loss assessment. Drug-induced alopecia is common, underdiagnosed, and potentially reversible — but only if identified. The mechanism may be direct telogen induction (shifting follicles prematurely into rest), anagen disruption (arresting matrix proliferation), androgenic stimulation, or inflammatory triggering. The latency between drug initiation and clinical manifestation varies by mechanism and is one reason drug causes are frequently missed.

Retinoids

Isotretinoin, acitretin — telogen effluvium-pattern shedding, typically dose-dependent; onset usually within weeks of initiation; may persist with continued use.

Anticoagulants

Heparin, warfarin, DOACs — telogen mechanism; shedding classically begins 8–16 weeks after initiation; often self-limiting once dose stabilised.

Anticonvulsants

Valproate in particular — associated with diffuse effluvium and, in some cases, patterned thinning; nutritional interaction through biotin metabolism may compound effect.

Antidepressants

SSRIs and SNRIs — reported but inconsistent association; may unmask or exacerbate existing susceptibility rather than acting as a direct cause; switching class may be diagnostically informative.

Weight Loss Agents and GLP-1 Agonists

Rapid weight loss — regardless of mechanism — is a powerful telogen effluvium trigger. Semaglutide and related agents are increasingly associated with effluvium in clinical practice, likely through the combined effect of rapid weight change and metabolic shift rather than direct drug action.

Hormonal Therapies and Anabolic Agents

Androgenic progestogens, testosterone replacement, anabolic steroids, and HRT formulations all carry potential to worsen AGA in susceptible individuals. This must be specifically asked about as patients frequently do not volunteer use of prescribed or non-prescribed anabolic agents.

The hair-specific treatment history should cover: current or previous minoxidil use (topical or oral, formulation, duration, compliance, and reason for cessation); anti-androgen therapy; PRP series; scalp mesotherapy; low-level laser or LED devices; previous hair transplantation; and any adverse reactions to scalp treatments. This history prevents repetition of failed approaches, identifies compliance failures that may have limited efficacy, and provides critical context for treatment planning.

4.6 Nutrition, Illness, and Trigger States

Telogen effluvium is triggered by any significant physiologic or metabolic stressor that disrupts the normal cycling balance of the scalp. The challenge in history-taking is identifying the correct trigger among multiple possible candidates — since patients frequently offer plausible-sounding but actually unsupported explanations for their shedding, particularly in the current era of social media-driven hair loss awareness.

Nutritional Triggers

Rapid weight loss, very low-calorie dieting, protein restriction, crash diets, restrictive eating disorders, and post-bariatric surgery states are all potent TE triggers. Key nutritional deficiencies include iron deficiency (serum ferritin, not haemoglobin alone), zinc, vitamin D, and protein. Assess dietary pattern, recent weight trajectory, and supplement use without endorsing patient assumptions about which nutrients are responsible.

Illness and Febrile States

Any significant febrile illness — including viral infections, bacterial sepsis, and COVID-19 — can induce acute TE with typical latency of 8–12 weeks from the febrile episode. The illness itself rather than the pathogen is the trigger. Degree of metabolic disruption, hospitalisation, and concurrent nutritional stress amplify the shedding response.

Surgery and Major Physical Stress

General anaesthesia and significant surgical procedures reliably trigger TE at 2–3 months post-operatively. Blood loss, nutritional depletion, and physiologic stress response all contribute. Shedding is typically self-limiting and resolvable once nutritional recovery is complete.

Psychological Stress

Severe acute psychological stress — bereavement, trauma, major life disruption — can trigger TE, but the association is weaker and less consistent than physiologic triggers. Patients frequently attribute hair loss to stress when a metabolic or nutritional cause is actually operative. Investigate both rather than accepting stress as an explanation without corroborating history.

Red Flag History Elements Requiring Urgent Evaluation: Scalp pain or burning; eyebrow loss or loss of body hair; rapidly enlarging patches; pustules, crusting, or ulceration; a personal or family history of autoimmune disease; concurrent systemic symptoms including joint pain, fatigue, or mucosal involvement; or any history suggesting traction or scarring processes. These features require a higher index of suspicion, accelerated workup, and often specialist referral. Do not reassure on the basis of clinical probability alone when red flags are present.

4.7 Psychosocial Impact and Consultation Style

Hair loss consultations sit at the intersection of clinical medicine and psychological medicine. The physician must simultaneously operate as a diagnostician, a scientist communicating evidence with appropriate calibration, and a clinician who understands that the patient's subjective experience of their hair loss may be profoundly distressing in ways that are entirely disproportionate to the objective severity of the condition. Both dimensions are clinically relevant and must be addressed.

The first principle is that psychological distress in the context of hair loss is real, clinically significant, and should be documented rather than minimised. Studies consistently demonstrate that hair loss — even when medically benign — is associated with reduced quality of life, body image disturbance, social withdrawal, depression, anxiety, and in some cases psychiatric morbidity. Dismissing distress as cosmetic triviality represents both a failure of empathy and a failure of clinical completeness.

The second principle is that distress must not drive inappropriate intervention. The physician's role includes containing distress through accurate prognostic information, realistic expectation-setting, and validation — not through offering more aggressive or less evidence-based treatments simply because the patient is distressed. The combination of psychological vulnerability and financial motivation to intervene makes this one of the most ethically demanding aspects of hair medicine practice.

Document the Burden

Record duration of concern, functional impact (avoidance, social withdrawal, professional impact), previous treatment history and its emotional as well as physical outcomes, and any stated psychiatric history or current mental health treatment. This is medically and legally relevant documentation.

Recalibrate Before Prescribing

A patient who has already cycled through multiple treatments with disappointment may need significant expectation recalibration before any new intervention is discussed. Introducing a new therapy prematurely to a patient with unrealistic expectations is likely to produce another cycle of disappointment regardless of the intervention's actual efficacy.

Refer When Appropriate

When psychological distress appears disproportionate, when body dysmorphic disorder is suspected, or when a patient's decision-making appears significantly impaired by emotional distress, a mental health referral — and a delay in elective procedural intervention — may be the most clinically appropriate response. Document the rationale carefully.

4.8 Structured Consultation Template

The following template provides a domain-by-domain consultation framework for use in clinical practice. It is designed to ensure diagnostic completeness, support hypothesis-driven examination, and provide documentation that meets medico-legal standards in a cosmetically sensitive clinical population. Each domain should be completed before the examination begins.

Consultation Domain	Questions to Ask	Why It Matters Diagnostically
Presenting Complaint	Is this shedding, thinning, breakage, recession, focal loss, or a combination? How does the patient describe what they see?	Defines the primary diagnostic branch and prevents premature diagnostic closure
Onset and Tempo	When did it start exactly? Was onset sudden or gradual? Is it getting worse, stable, or improving?	Distinguishes trigger states (TE) from progressive patterned disease; guides urgency of workup
Pattern and Symptoms	Where is loss most visible — crown, temples, diffuse, patchy? Any itch, pain, scale, burning, or visible scalp change?	Symptoms raise suspicion for inflammatory or scarring disease requiring urgent evaluation
Systemic and Endocrine History	Menstrual regularity, thyroid symptoms, weight change, acne, hirsutism, menopausal status (women); systemic diseases, metabolic history (all patients)	Identifies hormonal and systemic contributors that alter diagnosis, prognosis, and treatment selection
Triggers and Exposures	Recent illness, surgery, psychological stress, dietary change, medications initiated or stopped, postpartum state	Supports or refutes TE as a dominant or contributing process; identifies reversible causes
Drug and Treatment History	All current and recent medications; previous hair-specific treatments and their outcomes; adverse reactions; supplements	Identifies drug causes; prevents repetition of failed approaches; shapes treatment planning
Family History	Hair loss in first-degree relatives on both sides; age of onset; severity; PCOS or endocrine disorders in female relatives	Informs genetic susceptibility, prognosis, and long-term management strategy
Psychosocial Impact	Functional impact, duration of distress, prior treatment disappointment, self-image, social or professional consequences	Guides consultation style, expectation management, and identifies need for mental health support

4.9 End-of-Module Assessment Prompts

The following assessment prompts are designed to evaluate clinical reasoning and synthesis rather than factual recall. Responses should reflect structured thinking, appropriate use of diagnostic categories, and understanding of evidence quality. These prompts may be used for self-directed study, tutor-facilitated discussion, or as the basis for formal written assessments on the IIIOHR academy platform.

1

Differentiation by History

Using history alone — before examination or investigations — differentiate the likely diagnoses in a 28-year-old woman with sudden diffuse shedding, a 42-year-old man with progressive bitemporal recession, a 35-year-old woman with frontal thinning and irregular periods, and a 55-year-old woman with patchy loss and scalp burning. Explain the key historical features that drive each diagnosis.

2

Scalp Symptoms and the Differential

Explain why the presence of scalp symptoms — burning, pain, pruritus, or tenderness — materially alters the differential diagnosis in a patient presenting with hair loss. Which symptom patterns should heighten concern for inflammatory or scarring disease? What does their absence tell you and what does it not tell you?

3

Female Diffuse Thinning Workup

List the minimum endocrine and reproductive history required in any female patient presenting with diffuse hair thinning. Justify each domain and explain how the answer to each question alters the differential diagnosis or investigation plan. At what point does a non-contributory endocrine history in a woman change your working diagnosis?

4

Documenting Uncertainty

Describe in clinical terms how you would document diagnostic uncertainty without undermining patient trust or medico-legal clarity. Provide an example of language appropriate for a consultation note in a case where the differential includes both AGA and early lichen planopilaris, and a definitive distinction cannot yet be made on history and examination alone.

Module Summary: Clinical Trichology Foundations

Volume 1 of the Postgraduate Certificate in Clinical Trichology and Hair Restoration Medicine has established the foundational intellectual, biological, hormonal, and clinical architecture upon which all subsequent volumes will build. The following summary consolidates the core conceptual advances from each module.

Module 1: Scope and Standards

Physician-led trichology requires clinical reasoning, not pattern recognition alone. Evidence tiers govern all treatment communication. Ethical obligations include genuine informed consent, honest uncertainty documentation, and protection of vulnerable cosmetic populations from overpromising.

Module 2: Biological Architecture

The follicle is a dynamic signalling structure. The bulge stem-cell niche determines regenerative potential. Cycling biology explains clinical presentations. Scalp environment and ethnic variation are not peripheral considerations — they are clinically essential domains.

Module 3: Molecular Regulation

Hair loss is driven by interlocking pathways — androgen signalling, Wnt/beta-catenin, prostaglandins, growth mediators, microinflammation, and fibrosis. Treatment logic follows directly from pathway biology. No single intervention addresses the full molecular complexity of AGA.

Module 4: Clinical Consultation

A structured, hypothesis-driven history is the primary diagnostic instrument of the trichology consultation. Complaint categorisation, tempo analysis, endocrine history, red flag recognition, and psychosocial assessment are all non-negotiable components of a competent hair medicine consultation.

Key Principles: The Physician-Led Trichology Standard

> Clinical trichology, practised at the physician level, demands the same intellectual rigour, diagnostic precision, ethical discipline, and evidence-based thinking that characterises any other medical specialty. The complexity of the hair follicle and its disorders fully justifies this standard.

Reason Before Treating

Every treatment decision should be preceded by a completed differential diagnosis, an evidence-tier assessment of the proposed intervention, and documentation of the clinical basis for the chosen approach. Jumping from complaint to prescription is the single most common source of poor outcomes in hair medicine.

Biology Governs Logic

Understanding follicular biology — from stem-cell niches to molecular signalling cascades — is not academic supplementation. It is the mechanistic infrastructure that explains why treatments work, why they fail, why some patients respond and others do not, and why early intervention in progressive disease preserves more biological capital than late-stage rescue.

Ethics Is Clinical Competence

In a field with a high cosmetic dimension, psychological vulnerability in patients, and significant commercial pressure, ethical practice is not an afterthought. Honest communication of uncertainty, genuine informed consent, and refusal to overpromise are core clinical competencies — as important as diagnostic accuracy and pharmacologic knowledge.

Longitudinal Thinking

Hair medicine is a long-game specialty. Patients managed well at first consultation will have better outcomes over a decade than patients managed for immediate cosmetic response. Design every plan with five-year and ten-year implications in mind — donor stability, progression risk, treatment maintenance, and surgical eligibility must all be considered from the outset.

Clinical Red Flags: Quick Reference

The following features — identified in history, examination, or investigations — should elevate the clinical threshold and prompt further workup, specialist referral, or both. No single feature is absolutely pathognomonic, but their presence substantially changes the pre-test probability of serious, progressive, or irreversible disease.

History Red Flags

- Scalp pain, burning, or tenderness
- Eyebrow or eyelash loss
- Loss of body hair beyond scalp
- Rapidly enlarging patches over weeks
- Pustules, crusting, or ulceration on scalp
- Personal or family history of autoimmune disease
- Concurrent joint pain, fatigue, mucosal symptoms
- History of significant traction practices
- Systemic symptoms accompanying acute shedding
- Unexplained rapid progression despite treatment

Examination Red Flags

- Loss of follicular ostia on trichoscopy
- Perifollicular erythema, scaling, or casts
- Peripilar casts or tubular scaling
- Atrophic, smooth, or scarred scalp surface
- Pustulation or follicular plugging
- Asymmetric or bizarrely shaped patches
- Exclamation-mark hairs (alopecia areata)
- Broken hairs of uniform short length (trichotillomania)
- Marginal recession in a young woman (traction)
- Cadaverised hairs or black dots in patches

Investigation Red Flags

- Significantly elevated or suppressed serum ferritin
- Thyroid function abnormality
- Elevated free androgen index or testosterone
- Biopsy showing interface dermatitis or follicular destruction

Evidence Tier Reference Card

This reference card is designed for rapid clinical use. When formulating a treatment plan or communicating with patients, assign each proposed intervention to its appropriate evidence tier. Document the tier in the consultation note alongside the clinical rationale for recommending or discussing the intervention.

Tier 1 — Established Practice

Standard for clinical recommendation. Topical minoxidil (2% and 5%); oral finasteride in adult males with AGA (with reproductive counselling); dutasteride (Tier 1/2 depending on jurisdiction); standard diagnostic workup; scalp biopsy in suspected scarring alopecia; baseline investigations in female diffuse thinning.

Tier 2 — Evolving Practice

Discussable with evidence context. Low-dose oral minoxidil (0.25–5mg depending on sex and tolerability); spironolactone in female AGA with androgenic features; cyproterone acetate (where licensed); ketoconazole shampoo as adjunct; selected topical anti-androgen preparations.

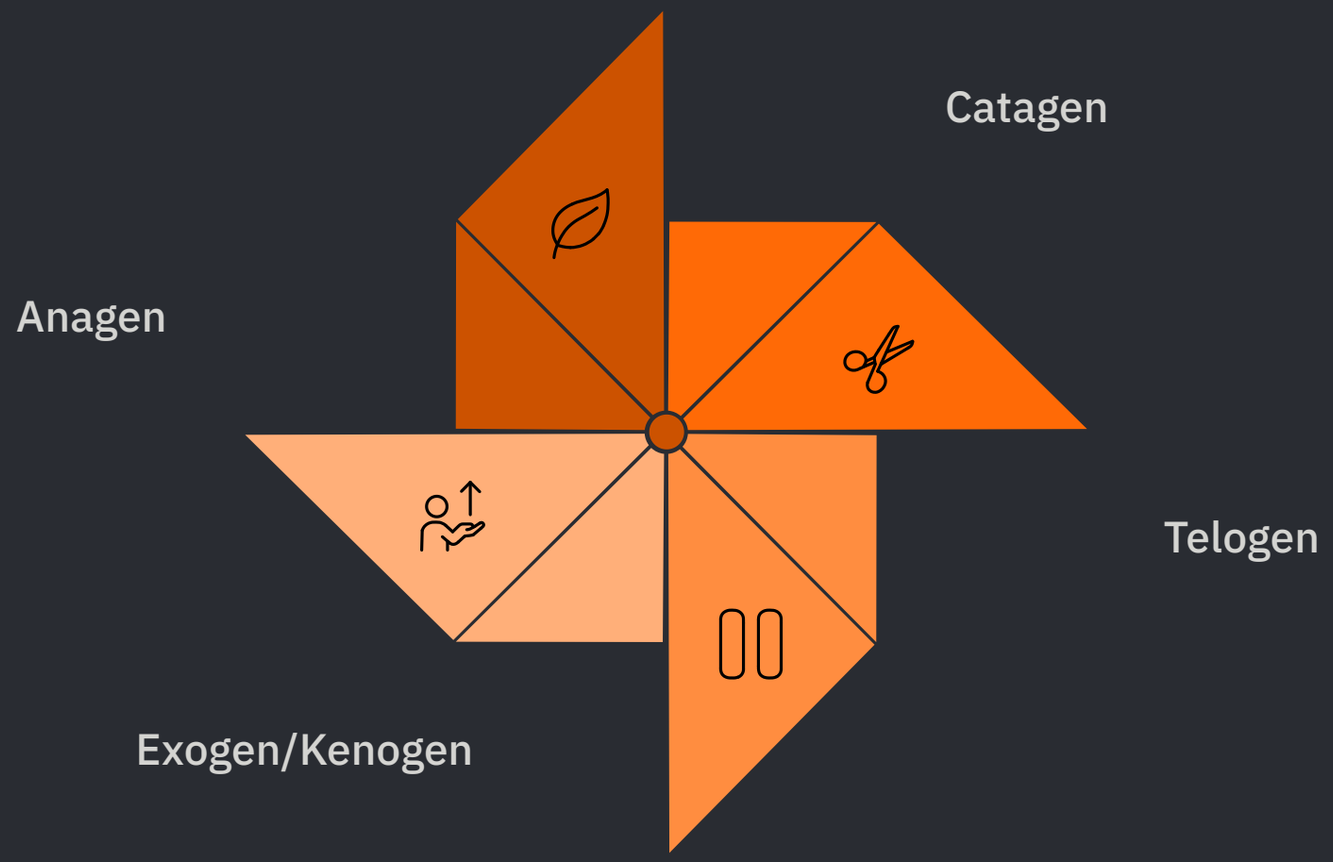
Tier 3 — Emerging Adjuncts

Requires clear heterogeneity communication. PRP (protocol-dependent outcomes); low-level laser/LED therapy; scalp microneedling as a penetration-enhancing adjunct; topical growth factor preparations; selected nutraceuticals with partial evidence.

Tier 4 — Investigational

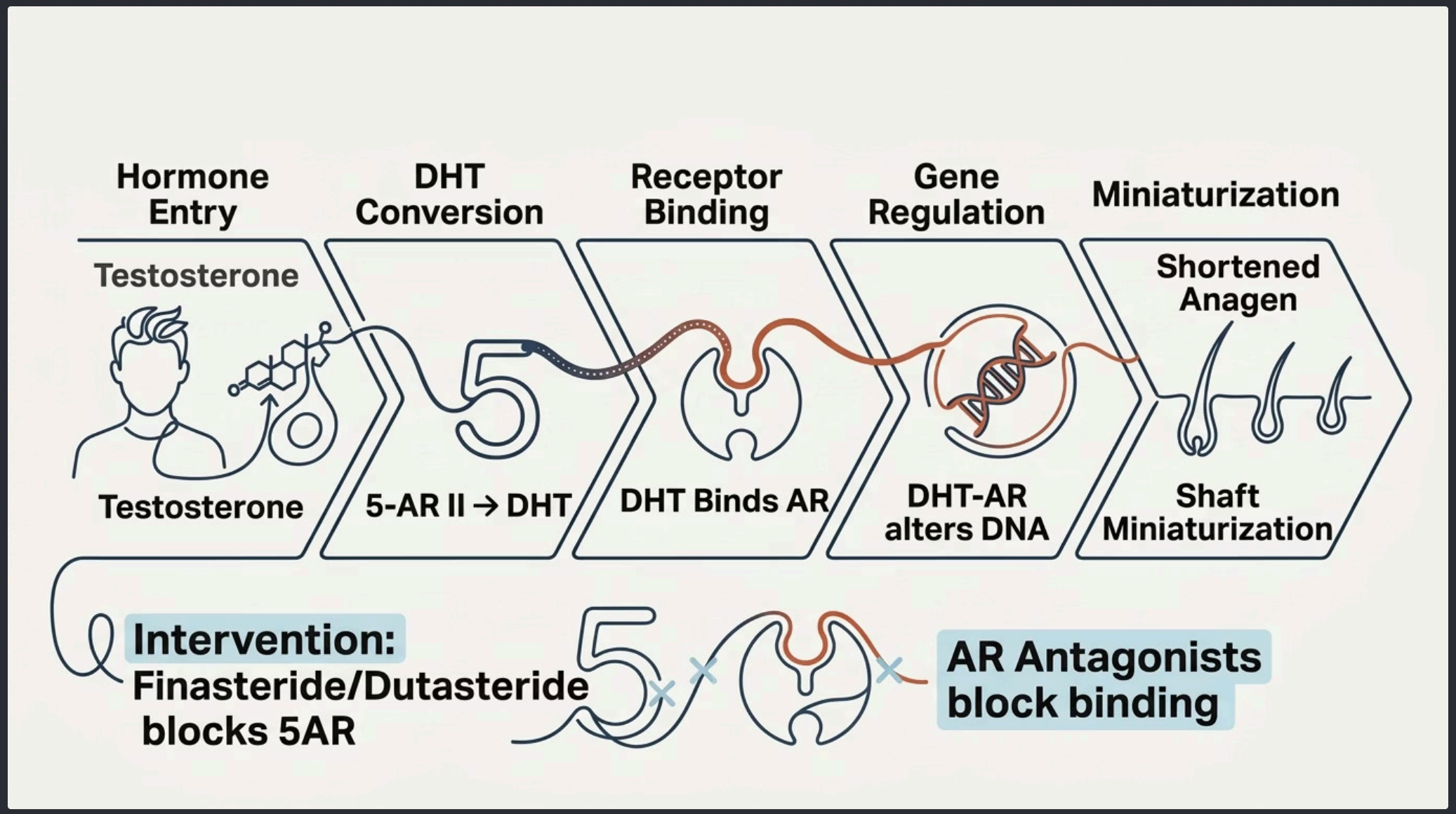
Biologically plausible, not guideline-standard. Exosome-based scalp therapy; partial transection grafting and regenerative donor concepts; stem-cell conditioned media preparations; Wnt pathway agonist topicals; JAK inhibitor use outside licensed alopecia areata indications.

Hair Cycle Phases: Clinical Integration Reference



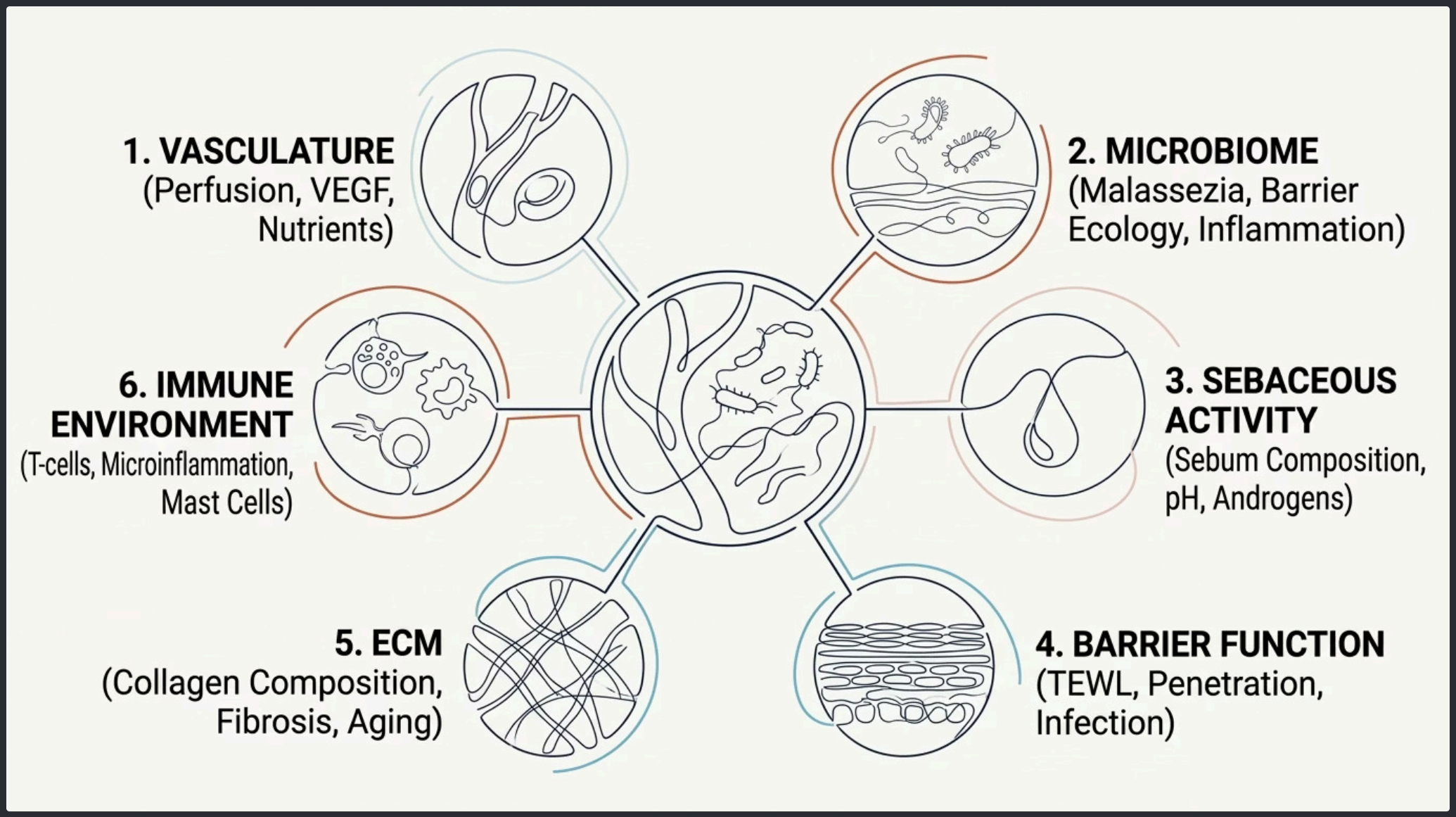
The clinical importance of cycle phase distribution cannot be overstated. In a healthy scalp, approximately 85–90% of follicles are in anagen at any given time, providing the high density and active growth that characterises a normal hair pattern. Disorders that shift even 10–15% of follicles into telogen produce clinically noticeable shedding. Disorders that progressively shorten successive anagen cycles — as in AGA — reduce both fibre calibre and apparent density over years. Understanding which phase is disrupted in a given disorder allows the physician to explain the mechanism of hair loss to patients in terms they can understand and to select interventions targeted at the appropriate phase of the cycle.

Androgen Pathway: Molecular to Clinical



This pathway schematic maps the principal molecular cascade responsible for androgenetic alopecia from circulating substrate to clinical hair loss. Each intervention point represents a therapeutic target. Finasteride and dutasteride act upstream at the conversion step. Androgen receptor antagonists (spironolactone, cyproterone acetate, bicalutamide) act at the receptor binding step. Understanding which step is targeted — and the degree to which downstream gene expression changes persist despite upstream blockade — explains why combination approaches may offer additive benefit and why neither class of agent produces complete reversal of established miniaturisation.

Scalp Microenvironment: A Systems View



The follicle does not exist in biological isolation. Its performance — cycling regularity, calibre maintenance, response to medical therapy, and surgical graft survival — depends on the quality of the surrounding scalp microenvironment. This systems view is clinically actionable: a physician who addresses only the follicle while neglecting vascular support, microbiome balance, barrier integrity, and perifollicular inflammatory load will achieve systematically inferior treatment responses compared to one who manages the full biological context. Volume 2 of this curriculum will explore examination findings and investigation strategies that allow clinical assessment of each of these environmental domains.

Looking Ahead: Volumes 2 and Beyond

Volume 1 has established the scientific and clinical foundations of physician-led trichology. The subsequent volumes in the Postgraduate Certificate curriculum build directly on these foundations, advancing from biological understanding to practical clinical examination, investigation, diagnosis, and management of the full spectrum of hair and scalp disorders encountered in practice.

01

Volume 2: Clinical Examination and Trichoscopy

Scalp and hair examination technique; trichoscopy pattern recognition across diagnoses; biopsy selection and interpretation; photography and documentation standards for longitudinal monitoring.

02

Volume 3: Differential Diagnosis and Disease Classification

Comprehensive classification of non-scarring and scarring alopecias; diagnostic algorithms for common and uncommon presentations; clinicopathologic correlation; referral logic and specialist interface.

03

Volume 4: Medical Management and Treatment Protocols

Evidence-based medical therapy across the evidence tiers; treatment sequencing and combination strategies; managing expectations; monitoring response; adverse effect management and switching protocols.

04

Volume 5: Hair Restoration Surgery and Procedural Medicine

Surgical candidacy assessment; FUT and FUE technique principles; donor planning; medical-surgical integration; regenerative adjuncts in surgical context; long-term surgical outcome management.

About the Hair Longevity Institute and IIIOHR

Hair Longevity Institute (HLI)

The Hair Longevity Institute was founded on the principle that hair medicine practised at the physician level should be scientifically rigorous, evidence-calibrated, and ethically grounded. HLI serves as the clinical and academic development arm of the curriculum, providing ongoing research integration, clinical case material, and expert faculty input to ensure the curriculum remains current with the evolving evidence base in trichology and hair restoration medicine.

HLI's longevity medicine framework situates hair health within the broader context of healthy aging, hormonal medicine, and preventive medicine — reflecting the reality that hair disorders in clinical practice rarely exist in isolation from systemic health considerations.

International Institute of Hair Restoration (IIIOHR)

The International Institute of Hair Restoration (IIIOHR) provides the accreditation framework, online learning platform, and international physician community through which the Postgraduate Certificate in Clinical Trichology and Hair Restoration Medicine is delivered. IIIOHR is committed to raising the global standard of physician-led hair medicine practice through structured postgraduate education, competency assessment, and peer community development.

The IIIOHR online academy is the digital home for all curriculum modules. Each module is structured for asynchronous study with integrated assessment, clinical case examples, and faculty-reviewed assignments that can be completed alongside active clinical practice.

Curriculum Governance

This curriculum is reviewed periodically by the HLI/IIIOHR Academic Board to incorporate emerging evidence, updated clinical guidelines, and feedback from the physician learner community. Evidence tier assignments for treatments discussed in this curriculum are subject to revision as new data become available.

Volume 1: End-of-Volume Assessment — Multiple Choice Questions

The following 20 questions assess knowledge across all four modules of Volume 1. Select the single best answer for each question. Answers are provided on the following page.

1. **1. Which of the following best defines the primary distinction between physician-led trichology and non-medical hair practice?**
   A. Use of topical treatments
   B. Access to dermoscopy equipment
   C. Diagnostic authority, prescribing rights, and ethical accountability
   D. Ability to perform hair transplantation
2. **2. Which evidence tier describes treatments supported by multiple randomised controlled trials and systematic reviews?**
   A. Tier 1
   B. Tier 2
   C. Tier 3
   D. Tier 4
3. **3. During embryologic follicle formation, which signalling interaction initiates placode formation?**
   A. Androgen receptor activation in mesenchyme
   B. Reciprocal Wnt signalling between epidermis and dermis
   C. Prostaglandin D2 release from keratinocytes
   D. VEGF-mediated vascular recruitment
4. **4. The bulge region of the hair follicle is located at which anatomical level?**
   A. Infundibulum, adjacent to the follicular ostium
   B. Isthmus, at the level of the arrector pili insertion
   C. Suprabulbar zone, above the dermal papilla
   D. Matrix, surrounding the dermal papilla
5. **5. Which phase of the hair cycle is characterised by active mitosis in matrix keratinocytes and full pigment production?**
   A. Catagen
   B. Telogen
   C. Anagen
   D. Exogen
6. **6. Approximately what percentage of scalp follicles are in anagen at any given time in a healthy adult?**
   A. 30–40%
   B. 50–60%
   C. 85–90%
   D. 95–100%
7. **7. 5-alpha reductase type II is most highly expressed in which follicular compartment?**
   A. Outer root sheath
   B. Dermal papilla
   C. Sebaceous gland
   D. Arrector pili muscle
8. **8. DHT binds the androgen receptor with what approximate affinity relative to testosterone?**
   A. Equal affinity
   B. Two to three times greater affinity
   C. Five times greater affinity
   D. Ten times greater affinity
9. **9. Which prostaglandin has been identified as an inhibitor of hair growth and is elevated in AGA-affected scalp?**
   A. PGE2
   B. PGF2α
   C. PGD2
   D. PGI2
10. **10. Wnt/beta-catenin signalling during the hair cycle is most critical for which transition?**
    A. Catagen to telogen
    B. Telogen to anagen
    C. Anagen to catagen
    D. Exogen to kenogen
11. **11. Perifollicular microinflammation in AGA is histopathologically characterised by which finding?**
    A. Neutrophilic infiltrate around the sebaceous gland
    B. Lymphocytic infiltrate and concentric fibrosis around the upper follicle
    C. Eosinophilic infiltrate in the dermal papilla
    D. Plasma cell infiltrate in the infundibulum
12. **12. Which of the following is the most appropriate first step when a patient presents with hair loss?**
    A. Order a full hormonal panel
    B. Precisely translate the patient's subjective complaint into a clinically defined category
    C. Perform a trichoscopy examination
    D. Prescribe a trial of minoxidil
13. **13. A patient reports hair shedding that began 3 months after a febrile illness. Which diagnosis is most consistent with this history?**
    A. Androgenetic alopecia
    B. Alopecia areata
    C. Telogen effluvium
    D. Anagen effluvium
14. **14. In women presenting with hair loss, which historical domain is considered central rather than optional?**
    A. Sleep history
    B. Menstrual, reproductive, and endocrine history
    C. Dietary supplement use
    D. Exercise frequency
15. **15. Which genetic locus carries the most clinically significant signal in AGA susceptibility?**
    A. TGFB1 on chromosome 19
    B. Androgen receptor gene on the X chromosome
    C. WNT10A on chromosome 2
    D. PTGDS on chromosome 9
16. **16. The hair shaft medulla is absent or discontinuous in which hair type?**
    A. Terminal scalp hair
    B. Beard hair
    C. Fine vellus hair
    D. Pubic hair
17. **17. Which of the following best describes the clinical significance of follicular senescence distinct from AGA?**
    A. It is caused by DHT-mediated miniaturisation
    B. It represents a progressive, age-related decline in cycling efficiency independent of androgen sensitivity
    C. It is reversible with finasteride therapy
    D. It affects only occipital follicles
18. **18. VEGF in the follicular context primarily supports which function?**
    A. Androgen receptor downregulation
    B. Perifollicular angiogenesis and nutrient delivery during anagen
    C. Stem cell quiescence in the bulge
    D. Sebum production in the infundibulum
19. **19. Which consultation domain is most likely to reveal a drug-induced cause of hair loss?**
    A. Family history
    B. Psychosocial impact assessment
    C. Drug and treatment history
    D. Nutritional history
20. **20. The structured consultation template in Module 4 is designed primarily to achieve which outcome?**
    A. Reduce consultation time to under 10 minutes
    B. Ensure diagnostic completeness and support hypothesis-driven examination
    C. Standardise prescribing decisions across all patients
    D. Replace the need for trichoscopy in routine practice

Volume 1: MCQ Answer Key

Correct answers with brief explanatory rationale for each question. Use these to consolidate understanding and identify areas for further review.

Q / Answer	Rationale
Q1 / C	Diagnostic authority, prescribing rights, and ethical accountability define the physician-led model. Equipment access and procedural skills are secondary to this foundational distinction.
Q2 / A	Tier 1 evidence represents the highest level: multiple RCTs, systematic reviews, and meta-analyses. Lower tiers reflect smaller studies, expert consensus, or mechanistic data only.
Q3 / B	Reciprocal Wnt signalling between epidermal keratinocytes and dermal mesenchyme initiates placode formation. This epithelial–mesenchymal dialogue is the foundational inductive event in folliculogenesis.
Q4 / B	The bulge is located in the isthmus at the level of the arrector pili insertion. It houses epithelial stem cells (EpSCs) critical for follicular regeneration.
Q5 / C	Anagen is the active growth phase, characterised by matrix keratinocyte mitosis, melanocyte activity, and full pigment production. It lasts 2–7 years on the scalp.
Q6 / C	Approximately 85–90% of scalp follicles are in anagen at any given time. This ratio underpins the clinical interpretation of hair counts and shedding rates.
Q7 / B	5-alpha reductase type II is most highly expressed in the dermal papilla of androgen-sensitive follicles, making this compartment the primary site of DHT production within the follicle.
Q8 / C	DHT binds the androgen receptor with approximately five times greater affinity than testosterone, explaining its disproportionate biological potency in susceptible follicles.
Q9 / C	PGD2 (prostaglandin D2) is elevated in AGA-affected scalp and acts as a hair growth inhibitor via the GPR44/CRTH2 receptor. PGE2 and PGF2α have pro-growth or neutral effects.
Q10 / B	Wnt/beta-catenin signalling is most critical for the telogen-to-anagen transition, driving stem cell activation and the initiation of a new growth cycle.
Q11 / B	Perifollicular microinflammation in AGA is characterised histopathologically by a lymphocytic infiltrate and concentric lamellar fibrosis around the upper follicle (infundibulum and isthmus).
Q12 / B	The first step is precise translation of the patient's complaint into a clinically defined category. Acting before this step (ordering tests, prescribing) risks diagnostic error.
Q13 / C	Telogen effluvium classically presents 2–4 months after a physiologic stressor such as febrile illness. The lag reflects the time for follicles to complete the telogen phase and shed.
Q14 / B	Menstrual, reproductive, and endocrine history is central in women with any hair loss presentation. It informs differential diagnosis, investigation strategy, and treatment selection.
Q15 / B	The androgen receptor gene on the X chromosome carries the strongest and most replicated genetic signal in AGA susceptibility across GWAS studies.
Q16 / C	Fine vellus hair characteristically lacks a medulla or has a discontinuous one. The medulla is a feature of terminal hair and is absent in the miniaturised fibres of AGA.
Q17 / B	Follicular senescence is a progressive, age-related decline in cycling efficiency that is independent of androgen sensitivity. It is not reversed by anti-androgen therapy.
Q18 / B	VEGF supports perifollicular angiogenesis and nutrient delivery during anagen. Reduced VEGF expression is associated with follicular miniaturisation and impaired growth.
Q19 / C	Drug and treatment history is the domain most likely to reveal drug-induced alopecia. Many causative agents are overlooked without systematic medication review.
Q20 / B	The structured consultation template is designed to ensure diagnostic completeness and support hypothesis-driven examination, not to standardise prescribing or replace clinical tools.